Cross Product

I. Definition

Vector Operation 3: Cross Product

Let $\mathbf{a} = [a_1, a_2, a_3]$, $\mathbf{b} = [b_1, b_2, b_3]$, and $\mathbf{n}$ is a unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

Then

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}|\,|\mathbf{b}| \sin\theta)\,\mathbf{n}$$

and so

$$|\mathbf{a} \times \mathbf{b}| =$$

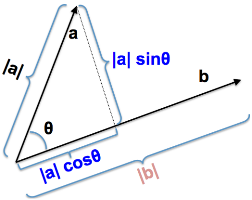
Notes:

- $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3 \quad \Longrightarrow$ and
- $\mathbf{a} \times \mathbf{b} \quad \mathbf{b} \times \mathbf{a}$
- A way to remember how to calculate a vector product is to use determinants:

$$\mathbf{a} \times \mathbf{b} =$$

II. Additional Concepts

Geometric Interpretation 1:



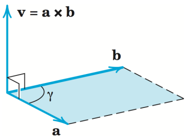
Dot Product:

$$\mathbf{a} \cdot \mathbf{b} = (\text{component of } \mathbf{a} \text{ along } \mathbf{b})(\text{length of } \mathbf{b})$$
$$= |a|\,|b| \cos\theta$$

Cross Product:

$$|\mathbf{a} \times \mathbf{b}| = (\text{component of } \mathbf{a} \text{ perpendicular to } \mathbf{b})(\text{length of } \mathbf{b})$$
$$= |a|\,|b| \sin\theta$$

Geometric Interpretation 2:



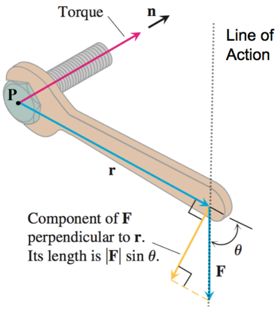
$|\mathbf{a} \times \mathbf{b}| =$

Properties: Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbf{R}^3$ and $d \in \mathbf{R}$ (*a scalar*)

- $d(\mathbf{a} \times \mathbf{b}) =$
- $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) =$
- $\mathbf{a} \times \mathbf{b} =$

III. Application

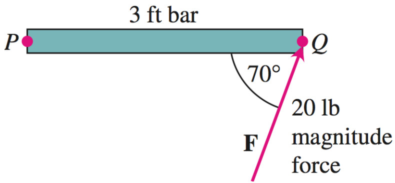
Application: Torque and Moment Force about an axis through point P.



Torque (Moment Force) is the tendency of a force to rotate an object about an axis.

$$Torque/Moment\ Force =$$

Verify that the torque vector in the diagram is pointing in the correct direction.



<u>Ex</u>: Find the Torque, τ, on an axis through P in the bottom diagram.

<u>Soln</u>:

$\tau =$

Therefore, $\tau =$